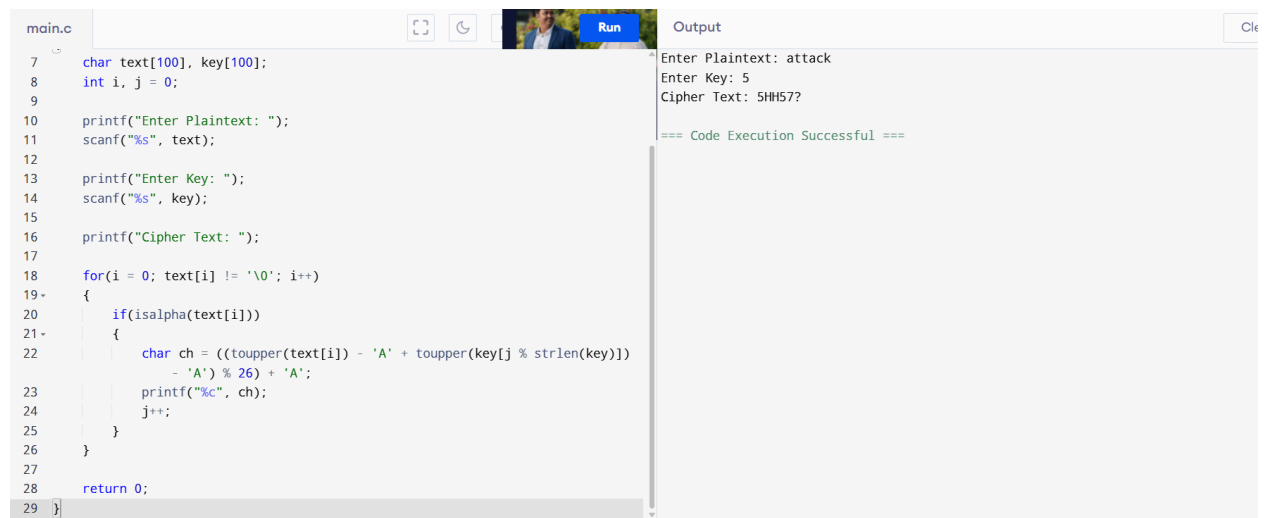


4. Vigenère (Polyalphabetic) Cipher



The image shows a code editor window with a file named 'main.c'. The code implements a Vigenère cipher. It prompts the user to enter a plaintext and a key. The plaintext 'attack' and key '5' are entered. The resulting ciphertext is '5H#57?'. The code uses the formula $ch = ((\text{toupper}(\text{text}[i]) - 'A' + \text{toupper}(\text{key}[j \% \text{strlen}(\text{key})]) - 'A') \% 26) + 'A'$ to calculate the ciphertext. The output window shows the execution results and a success message.

```
main.c
7 char text[100], key[100];
8 int i, j = 0;
9
10 printf("Enter Plaintext: ");
11 scanf("%s", text);
12
13 printf("Enter Key: ");
14 scanf("%s", key);
15
16 printf("Cipher Text: ");
17
18 for(i = 0; text[i] != '\0'; i++)
19 {
20     if(isalpha(text[i]))
21     {
22         char ch = (((toupper(text[i]) - 'A' + toupper(key[j % strlen(key)])
23                     - 'A') % 26) + 'A');
24         printf("%c", ch);
25         j++;
26     }
27 }
28 return 0;
29 }
```

Output

Enter Plaintext: attack
Enter Key: 5
Cipher Text: 5H#57?
=== Code Execution Successful ===